

Many-Testing Lower Bounds for Low-Latency Privacy on Hierarchical Overlays

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Abstract

We formalize a global-surveillance failure mode for low-latency routed systems: **many-testing**. A global passive adversary (GPA) can evaluate a large family of candidate witnesses (prefix regions, cuts, bins, edges) and win by selecting whichever witness is accidentally “clean” under cover traffic. Even when any fixed witness is rare, a best-of- B scan forces *logarithmic* overhead to suppress the witness across the entire family.

We give sharp emptiness / low-count thresholds (Poissonized and fixed-total), and a dependence-robust theorem that lower-bounds scan success using a packing number under overlap. We instantiate the framework on hierarchical (prefix-partitioned) overlays to obtain a canonical cover-rate floor under a deadline window of length L : for typical $B \approx n$ candidate prefix regions, the cover rate must satisfy $\lambda \gtrsim \frac{1}{L} \ln(n/\varepsilon)$ to make scan distinguishers fail with probability at most ε . Finally, we show that when edges are observable, hiding the *neighbor* choice requires per-edge (not just per-node) equalization, yielding an $\Omega(\log^2 n/L)$ -type floor in classic DHT scaling.

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1 Introduction and contributions

The guiding phenomenon is simple:

- A GPA can test many candidate “places” where a protocol’s behavior might become visible (prefix regions, edges, bins, committees).
- If a crisp rare event (like an empty baseline count) serves as a witness, then the GPA’s best-of-many advantage forces logarithmic overhead to suppress that witness everywhere.

This paper is intentionally proof-oriented. It is not a new protocol; it is a **lower-bound lens** that explains why several natural defenses “hit a floor” in low-latency regimes.

Main contributions (informal):

- **Sharp many-testing thresholds.** Exact emptiness/low-count scan bounds under Poissonization and fixed totals, capturing the GPA’s best-of- B advantage.
- **Overlap-robust scans.** A packing-number theorem that lower-bounds many-testing success even when candidate witnesses overlap.
- **Hierarchical-overlay instantiation.** Applying the framework to prefix regions yields the canonical low-latency cover floor $\lambda \gtrsim \frac{1}{L} \ln(n/\varepsilon)$ (in typical $B \approx n$ regimes).
- **Edge observability floor.** If the transcript reveals which neighbor edge is used, then per-edge equalization is necessary, giving an $\Omega(\log^2 n/L)$ -type floor in classic DHT scaling.

Attack-as-distinguisher viewpoint. The many-testing phenomenon is not just an existential bound: it yields an explicit transcript-level distinguisher. the low-count scan scans a tested family (e.g., all depth- k prefix regions) and flags a lookup iff some region has an unusually small observed total. Lemma 4.2 gives closed-form false-positive and true-positive bounds, and Proposition 4.1 translates them into the canonical cover-rate floor by substituting $\mu = n \lambda L / B$.

Threat model and observables

We consider a **global passive adversary (GPA)** who observes a transcript over a fixed time window of length L rounds (or an equivalent bounded-latency window).

Test family. Fix a finite family of “places” that can be scanned in the transcript: $I = \{1, 2, \dots, B\}$. Each $i \in I$ corresponds to some region / cut / bin / committee / edge-event predicate S_i (e.g., a prefix region in a hierarchical overlay). The GPA observes a nonnegative integer statistic C_i for each i , typically “how many messages crossed / were directed to / were attributed to S_i ” inside the window.

Null model (baseline cover). Under pure cover traffic, we model C_i as approximately Poisson with mean μ_i , often specialized to a common mean μ . The many-testing theorems in §2 use exactly Poisson (or fixed-total occupancy) to get sharp thresholds; §3 explains how overlap or local dependence affects constants rather than the log-law.

Alternative model (forced entry). Under a correctness requirement, the protocol forces bounded “signature touches” in a small number of tested regions early in the window. Formally, we assume there is a target-dependent index $i_\star$ such that, with failure probability at most η , an additional nonnegative increment $F_{i_\star} \in \{1, 2, \dots, m\}$ is added to the baseline count: $C_{i_\star} = X_{i_\star} + F_{i_\star}$, where $X_{i_\star}$ is the baseline component. The adversary does **not** need to separate X and F ; all distinguishers we analyze use only the observable totals $\{C_i\}$.

This formulation captures the common situation where “baseline = 0” is unobservable but “total is unusually small” (e.g., $C_{\{i\star\}} \leq m$) is observable and still yields a witness.

2 Setting and observables

We consider a GPA observing metadata at the network layer. The weakest model we analyze is count-only per observed unit (e.g., per directed edge per round). Stronger models (packet traces with timing/order) can only help the adversary.

Tested families and transcript counts

Let I index a family of tests. Each test $i \in I$ corresponds to a region / cut / bin / edge whose **total observed count** C_i over a fixed observation window of L rounds is available to the GPA (e.g., “# packets entering prefix region S ”, “# uses of edge e ”, “# contacts to bin (c,b) ”).

For analysis it is useful to decompose $C_i = B_i + W_i$, where B_i is **baseline cover** (workload-independent) traffic and W_i is **workload-dependent** traffic (real lookups, responses, etc.). The GPA observes C_i but does not (by design) label which packets are baseline vs workload.

A canonical witness in many-testing arguments is **baseline emptiness**: $E_i := \{B_i = 0\}$. E_i is generally not directly observable from C_i . However, correctness can force that some target-dependent test $i\star$ receives a small, bounded number of workload packets within the window.

Define the **observable low-count witness** (parameter $m \geq 1$): $Li(m) := \{1 \leq C_i \leq m\}$. This is measurable from the transcript whenever C_i is.

Lemma 1.1 (Observable low-count witnesses dominate baseline emptiness). Fix i and write $C_i = B_i + W_i$ with B_i, W_i nonnegative integers. 1) If $W_i \geq 1$ then $Li(m)$ implies $B_i \leq m-1$. 2) If $1 \leq W_i \leq m$ then $E_i \wedge \{W_i \geq 1\}$ implies $Li(m)$. 3) If $W_i = w_0$ deterministically for some $1 \leq w_0 \leq m$, then $\{C_i = w_0\}$ is equivalent to E_i and is therefore an observable surrogate for baseline emptiness.

Proof: (1) If $C_i \leq m$ and $W_i \geq 1$ then $B_i = C_i - W_i \leq m-1$. (2) If $B_i=0$ and $1 \leq W_i \leq m$ then $C_i = W_i \in [1, m]$. (3) If $W_i = w_0$ then $C_i = w_0$ iff $B_i = 0$. $\square$

We will use Theorem 2.3 below to quantify how large μ must be to suppress low-count witnesses $Li(m)$ across many tests.

3 Emptiness thresholds

Setup. Consider B candidate witnesses/tests. Under baseline-only traffic, each test produces an independent count $C_i \sim \text{Pois}(\mu)$. For an integer $m \geq 0$, define the low-tail probability

$$p_m(\mu) := \Pr[\text{Pois}(\mu) \leq m] = e^{-\mu} \sum_{j=0}^m \frac{\mu^j}{j!}.$$

Theorem 2.3 (Many-testing threshold for low counts). Let $E_m := \{\min_{i \in [B]} C_i \leq m\}$ be the *low-count scan* event. Then

$$\Pr[E_m] = 1 - (1 - p_m(\mu))^B.$$

Consequently, requiring $\Pr[E_m] \leq \varepsilon$ forces (and is achieved up to lower-order terms by)

$$\mu \geq \ln\left(\frac{B}{\varepsilon}\right) + m \ln \ln\left(\frac{B}{\varepsilon}\right) - \ln(m!) - O(1).$$

In particular, the emptiness case $m = 0$ yields the sharp logarithmic floor $\mu \geq \ln(B/\varepsilon) - O(1)$.

Proof sketch. The exact formula is by independence. For the threshold, for $\mu \geq 2m$ the Poisson low tail satisfies the two-sided bound

$$\frac{1}{2} e^{-\mu} \frac{\mu^m}{m!} \leq p_m(\mu) \leq 2 e^{-\mu} \frac{\mu^m}{m!},$$

so in the rare-witness regime $\Pr[E_m] \approx B p_m(\mu)$ and inverting $B p_m(\mu) \lesssim \varepsilon$ gives the stated expression.

Corollary 2.4 (*r*-hit low-count scans). Let $Z := |\{i : C_i \leq m\}| \sim \text{Bin}(B, p_m(\mu))$ and define $E_{m,r} := \{Z \geq r\}$. In the rare-witness regime $t := B p_m(\mu) \leq 1$ and $B \geq 2r$,

$$\Pr[E_{m,r}] = \Theta\left(\frac{t^r}{r!}\right),$$

so $\Pr[E_{m,r}] \leq \varepsilon$ implies $B p_m(\mu) \lesssim (\varepsilon r!)^{1/r}$, equivalently

$$\mu \geq \ln\left(\frac{B}{\varepsilon^{1/r}}\right) + m \ln \ln\left(\frac{B}{\varepsilon^{1/r}}\right) - \ln(m!) - O(1).$$

Remark (fixed-total occupancy). If a window contains exactly M baseline events thrown uniformly into B tests, the same logarithmic floor holds with $\mu \approx M/B$; e.g., suppressing emptiness with probability at most ε requires $M \geq B \ln(B/\varepsilon) - O(B)$.

4 Dependency-robust many-testing via packing and local dependence

Many natural test families overlap (nested prefix cuts, overlapping windows). We isolate an effective number of independent tests.

Poisson-atom baseline and packing number

Let Ω be a set of independent Poisson atoms: for each $a \in \Omega$, $N_a \sim \text{Pois}(\mu_a)$ independent. For each test i choose an atom set $A_i \subseteq \Omega$ and define $B_i := \sum_{a \in A_i} N_a$, $\lambda_i := \mathbb{E}[B_i] = \sum_{a \in A_i} \mu_a$. Then $B_i \sim \text{Pois}(\lambda_i)$. If $A_i \cap A_j = \emptyset$ then B_i and B_j are independent.

Define the disjoint packing number: $\Pi(\{A_i\}) := \max\{|J| : J \subseteq I \text{ and } A_i \cap A_j = \emptyset \text{ for all } i \neq j \text{ in } J\}$.

Theorem 3.1 (Packing-number many-testing bound). Let $\lambda_{\max} = \max_i \lambda_i$. Then $\Pr[\exists i : B_i = 0] \geq 1 - (1 - e^{-\lambda_{\max}})^{\Pi}$. In particular, enforcing $\Pr[\exists i : B_i = 0] \leq \varepsilon$ implies $\lambda_{\max} \geq \ln(\Pi/\varepsilon) + O(1)$.

Proof: choose a disjoint subfamily J of size Π ; on J , $\Pr[B_i = 0] \geq e^{-\lambda_{\max}}$ and events are independent. $\square$

Remark (tools and examples). The packing reduction can be phrased via dependency graphs and Chen–Stein Poisson approximation; worked overlap examples and a broader “minimal test” discussion are omitted here for brevity.

5 Prefix partitions and the $\Omega(\ln n / L)$ cover floor (toy baseline)

This section instantiates the many-testing lower bound on a stylized hierarchical overlay in a *three-step pipeline*: (i) identify a large family of target-dependent regions that can be tested from the transcript; (ii) argue that a correct lookup forces a small, *bounded* amount of workload traffic into *one* such region; (iii) apply the many-testing threshold to conclude a cover-rate floor.

Step 1: a size- B prefix family. Nodes have b -bit identifiers. For a target t , define the depth- k prefix region $S_k(t)$ as the set of nodes whose first k bits match t . The family $\mathcal{F}_k := \{S_k(u)\}_{u \in \{0,1\}^k}$ has size $B = 2^k$, and for random IDs each region has size about n/B .

Toy baseline (UCR). Uniform Constant Rate (UCR) baseline: each node emits λ baseline packets per round to a uniform random destination. For a region S ,

$$\mu(S) := \mathbb{E}[\text{baseline ingress into } S \text{ over } L \text{ rounds}] = \lambda L |S| (n - |S|) / (n - 1),$$

so for balanced depth- k regions ($|S| \approx n/B$ and $B \ll n$), $\mu(S) \approx \lambda L n / B$. Applying Theorem 2.1 across B disjoint tests already suggests the canonical scale: to suppress best-of- B emptiness with failure probability $\leq \varepsilon$, one needs $\mu \gtrsim \ln(B/\varepsilon)$, i.e. $\lambda \gtrsim (B/(nL)) \ln(B/\varepsilon)$, and in the deepest regime $B \approx n$ this becomes $\lambda \gtrsim (1/L) \ln(n/\varepsilon)$.

Step 2: a transcript-measurable forced-entry witness. Let $C(S)$ be the **total observed** number of packets entering S in the L -round window, and write $C(S) = B(S) + W(S)$ where $B(S)$ is baseline cover traffic and $W(S)$ is workload-dependent traffic.

Assumption 4.1 (Forced entry with bounded signature). For the true target t , with probability at least $1 - \eta$ there exists a depth- k region $S_k(t)$ such that $1 \leq W(S_k(t)) \leq m$ for some small fixed m , and the GPA observes only the totals $\{C(S)\}$ (so events like $\{C(S_k(t)) \leq m\}$ are transcript-measurable).

Step 3: one scan distinguisher. **Definition 4.1 (Low-count scan).** Given the transcript, compute the tested family $\mathcal{F}$ (e.g., all depth- k prefix regions) and output “forced entry” iff $\min_{S \in \mathcal{F}} C(S) \leq m$.

Lemma 4.2 (Low-count scan advantage). Let $\mathcal{F}$ be a family of B tests whose baseline ingress counts are effectively independent with $B(S) \sim \text{Pois}(\mu)$ under H_0 (baseline only). Under H_1 (forced entry) assume Assumption 4.1 holds. Let $p_j(\mu) := \Pr[\text{Pois}(\mu) \leq j]$. Then the low-count scan has

$$P_0 = 1 - (1 - p_m(\mu))^B, \quad P_1 \geq (1 - \eta) p_{m-1}(\mu).$$

Proof sketch. Under H_0 , the scan triggers iff at least one of the B tests has $C(S) \leq m$. Under H_1 , on the forced-entry event there is a region with $1 \leq W(S) \leq m$, and whenever $B(S) \leq m - 1$ we have $C(S) = B(S) + W(S) \leq m$.

Proposition 4.1 (Canonical cover floor from forced entry via low counts). Assume there are B effectively independent depth- k regions whose baseline ingress counts are i.i.d. $\text{Pois}(\mu)$ with $\mu \approx n\lambda L / B$ in UCR. Suppose Assumption 4.1 holds and we require that a GPA’s scan succeeds with probability at most ε under baseline-only traffic. Then necessarily

$$\mu \geq \ln(B/\varepsilon) + O(1),$$

and more sharply (for fixed m) $\mu \geq \ln(B/\varepsilon) + m \ln \ln(B/\varepsilon) - \ln(m!) - O(1)$ as in Theorem 2.3. Substituting $\mu \approx n\lambda L/B$ yields the cover-rate floor $\lambda \gtrsim (B/(nL)) \ln(B/\varepsilon)$, and in the deepest regime $B \approx n$ this reduces to $\lambda \gtrsim (1/L) \ln(n/\varepsilon)$.

Why Assumption 4.1 is natural (idealized). In a stylized greedy step, if a node chooses the best of m candidate contacts whose IDs are uniform among those matching the current prefix, the additional prefix gain G satisfies $\Pr[G \geq r] = 1 - (1 - 2^{-r})^m$ and $\mathbb{E}[G] \in [\log_2 m, \log_2 m + 1]$. This “best-of- m buys prefix bits” heuristic explains why hierarchical lookups tend to inject a small, target-dependent signature into progressively deeper prefix regions.

6 Edge observability and per-edge scheduling floors

The many-testing arguments above focus on *counts over a family of tests*. A different limitation applies when the GPA observes traffic *per edge* (or per neighbor) and the defense attempts to hide which edges are “active” by per-edge padding.

A basic per-edge latency–rate tradeoff

Consider a route that traverses h directed edges. Suppose each edge e emits cover packets according to a schedule with rate ρ (packets per round). In the strongest “edge-observable” model, the GPA sees per-edge emission times (or equivalently, counts in each round per edge).

A minimal requirement for low latency is: a real payload packet should traverse the h -hop path within L rounds with high probability. If the schedule is the only way to move packets (i.e., payload rides on scheduled emissions), then each hop incurs an expected waiting time at least $1/\rho$.

Lemma 7.1 (Edge scheduling floor; rate grows with path length). In the per-edge scheduled-emission model above, any scheme that guarantees expected end-to-end latency at most L on an h -hop path must satisfy $\rho \geq h/L$ for every edge on that path. Consequently, for a node of out-degree Δ , the per-node scheduled-slot rate must be at least $\Delta\rho = \Omega(\Delta h / L)$.

Proof. Under any stationary per-edge schedule of rate ρ , the expected time until the next emission on an edge is at least $1/\rho$ (e.g., by renewal theory; for a Poisson clock it is exactly $1/\rho$, and for deterministic periodic schedules it is also $1/\rho$). A packet traversing h edges therefore has expected latency at least $h \cdot (1/\rho)$. Requiring $h/\rho \leq L$ yields $\rho \geq h/L$. Summing over Δ outgoing edges gives the per-node rate lower bound. $\square$

Classic DHT scaling: $\Omega(\log^2 n / L)$

In many overlays, $h = \Theta(\log n)$ hops and $\Delta = \Theta(\log n)$ outgoing neighbors are typical scales (e.g., logarithmic-degree routing tables and logarithmic route length).

Corollary 7.2 (Edge-observability floor in logarithmic overlays). If $\Delta = \Theta(\log n)$ and $h = \Theta(\log n)$, then any per-edge schedule that maintains expected latency $\leq L$ requires per-node scheduled-slot rate $\Omega(\log^2 n / L)$.

This edge-observability floor is orthogonal to the many-testing and floors: it applies even when there is only a single tested region, as long as the GPA can observe traffic per edge.

7 Synthesis: what the lower bounds force

The results above can be read as a short design checklist for low-latency metadata-hiding lookups.

- **Best-of-many scans impose logarithmic mass.** If a GPA can scan an effective family of size B , then suppressing emptiness/low-count witnesses across the family forces baseline mass $\mu = \Omega(\ln(B/\varepsilon))$ in the relevant regions.
- **Hierarchical structure turns B into n .** In prefix-partitioned overlays, natural candidate families have size comparable to the population ($B \approx n$), yielding the canonical cover floor $\lambda \gtrsim \frac{1}{L} \ln(n/\varepsilon)$ in deadline L .
- **Observable edges require per-edge scheduling.** When the transcript reveals the neighbor edge used, hiding the choice requires per-edge (not just per-node) equalization, producing additional degree-dependent floors (e.g., $\Omega(\log^2 n/L)$ under classic DHT scaling).

These statements are intentionally protocol-agnostic: they describe what *any* defense must pay in order to make the corresponding scan distinguishers fail.

8 Related work and positioning

Our goal is not to propose a full protocol but to isolate *proof-friendly, transcript-measurable witnesses* that a global passive adversary (GPA) can test at scale, and to show that suppressing those witnesses imposes logarithmic (or worse) baseline overhead in low-latency regimes.

Overlay motivation. The routing picture is inspired by XOR-metric DHT overlays such as Kademlia [MM02] and by anonymous DHT lookup constructions that attempt to hide query targets using dummy traffic and/or split paths (e.g., Octopus [WB12]). These systems naturally induce large families of “regions” (prefix buckets, routing cuts) that can be scanned for rare events such as *no contact* or *few contacts*.

Low-latency anonymity motivation. On the communication-systems side, low-latency anonymity work highlights the tension between rate equalization and global traffic analysis, e.g., Loopix-style Poisson mixing with cover traffic [PH+17], round-based private communication systems such as Pung [A+16], and distributed metadata-private messaging such as Stadium [T+17]. Our contribution is complementary: even if a design equalizes *marginals* (per-node or per-bin), a GPA can often test *many* regions/bins/edges and win by selecting whichever produces a crisp rare event.

Probabilistic backbone. The many-testing phenomena we use are classical in Poisson/occupancy asymptotics and Poisson approximation: Le Cam’s Poisson-binomial approximation [LC60] (and expositions such as [Ste94]), dependency-graph Chen–Stein bounds in the style of Arratia–Goldstein–Gordon [AGG89], and large-deviation Poisson approximation heuristics (e.g., [Jan90]). We package these tools in a way that is easy to instantiate for hierarchical overlays and low-latency privacy designs.

Positioning

A recurring theme is that “equalize the marginal” (per-node rate, per-bin usage, per-edge slots) is not enough when the GPA can test *many* places and look for a crisp, rare event. Our contribution is to turn that intuition into referee-checkable inequalities with minimal protocol detail (isolated in explicit assumptions like Assumption 4.1).

9 Limitations and open problems

- **Transcript model.** We focus on GPA-visible count statistics over a fixed window L . Extending the scan framework to richer observables (timing, correlation, queueing) is important.
- **Effective test size.** The parameter B is application-dependent; formalizing *which* candidate families a real adversary can scan (and how dependent they are) remains case-specific.
- **Protocol specificity.** Our instantiations are stylized (prefix regions, edge choices). Sharper constants and protocol-specific bounds require matching the witness family to an implementation.
- **Achievability.** Designing defenses that match these floors under realistic constraints (bandwidth caps, churn, partial observability) is still largely open.

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